LAPORAN PRAKTIKUM

IOT 2

TUGAS Module 4 & Module 5



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TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

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1. Tambahkan variabel LED PWM

Praktikum_1.ino

```
#include <ESP8266WiFi.h>
#include <ESPAsyncTCP.h>
#include <ESPAsyncWebServer.h>
#include <DHT.h>

const char* ssid = "V2030";
const char* password = "raid_0507";

// Konfigurasi Pin
const byte dhtPin = 2;         // D4 (GPIO 2)
const byte buttonPin = 4;      // D2 (GPIO 4)
const byte ledPin = 12;        // D6 (GPIO 12)
const byte pwmLedPin = 5;  // D1 (GPIO5)
int pwmValue = 0;

DHT dht(dhtPin, DHT22);

// Variabel Pelacak Status (State & Cache)
bool ledState = false;
String currentTemp = "--";
```

2. Ubah bagian setup()

Praktikum_1.ino

```
        break;
    }
}

void setup() {
  Serial.begin(115200);

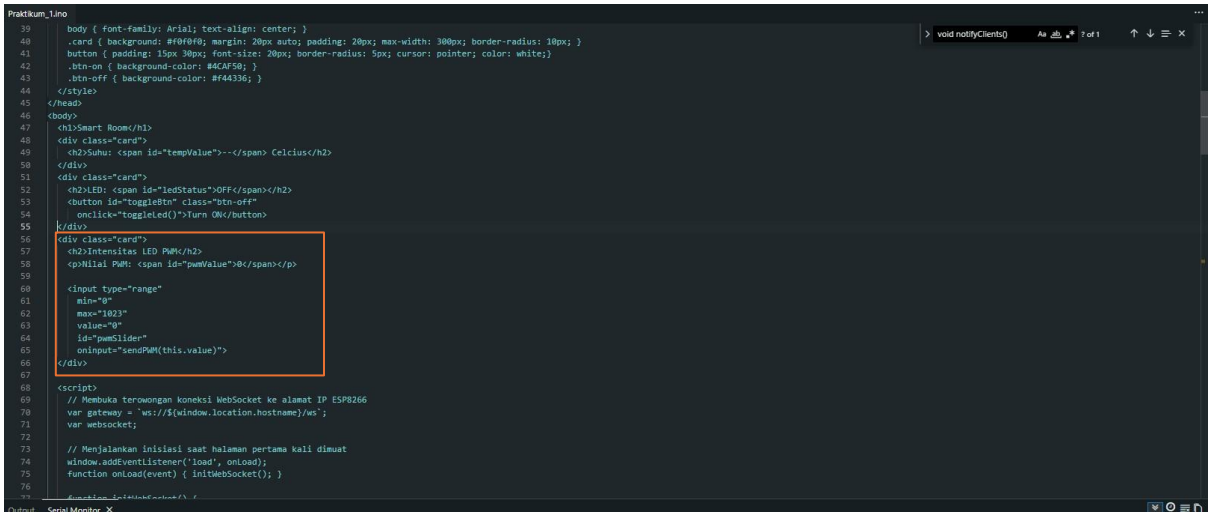
  // Mengatur pin tombol sebagai input (dengan resistor pull-down eksternal)
  pinMode(buttonPin, INPUT);

  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);

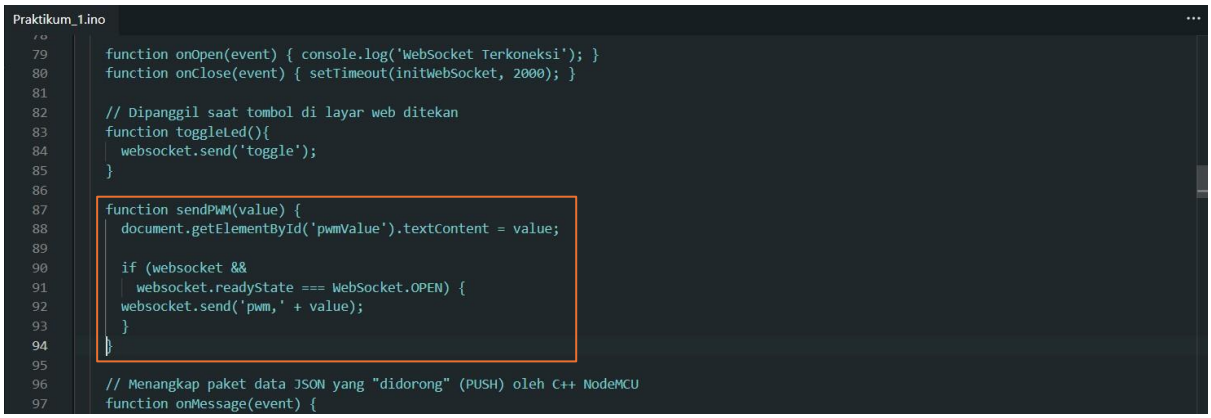
  pinMode(pwmLedPin, OUTPUT);
  analogWriteRange(1023);
  analogWrite(pwmLedPin, 0);

dht.begin();
```

3. Tambahkan slider pada HTML



4. Tambahkan fungsi JavaScript



5. Ubah fungsi notifyClients()

Praktikum_1.ino

```
</body>
</html>
)rawliteral";

// ---------------- BACK-END & WEBSOCKET LOGIC ----------------

// Fungsi menyebarkan paket data JSON ke SELURUH browser pengunjung
void notifyClients() {
  String jsonString = "{\"led\":\"";
  jsonString += String(ledState ? 1 : 0);
  jsonString += "\",\"suhu\":\"";
  jsonString += currentTemp;
  jsonString += "\",\"pwm\":\"";
  jsonString += String(pwmValue);
  jsonString += "\"}";

  ws.textAll(jsonString);
}
```

6. Tambahkan penanganan pesan PWM

```
void handleWebSocketMessage(
  void *arg, uint8_t *data, size_t len
) {
  AwsFrameInfo *info = (AwsFrameInfo*)arg;

  if (!(info->final && info->index == 0 &&
        info->len == len && info->opcode == WS_TEXT)) {
    return;
  }

  // Menyalin pesan ke buffer yang aman
  char message[32];

  if (len == 0 || len >= sizeof(message)) {
    return;
  }

  memcpy(message, data, len);
  message[len] = '\0';

  // ===== KONTROL LED UTAMA =====
  if (strcmp(message, "toggle") == 0) {
    ledState = !ledState;
    digitalWrite(ledPin, ledState ? HIGH : LOW);
    notifyClients();
    return;
  }

  // ===== KONTROL LED PWM =====
  if (strncmp(message, "pwm,", 4) == 0) {
    char *endPtr;
    long value = strtol(message + 4, &endPtr, 10);

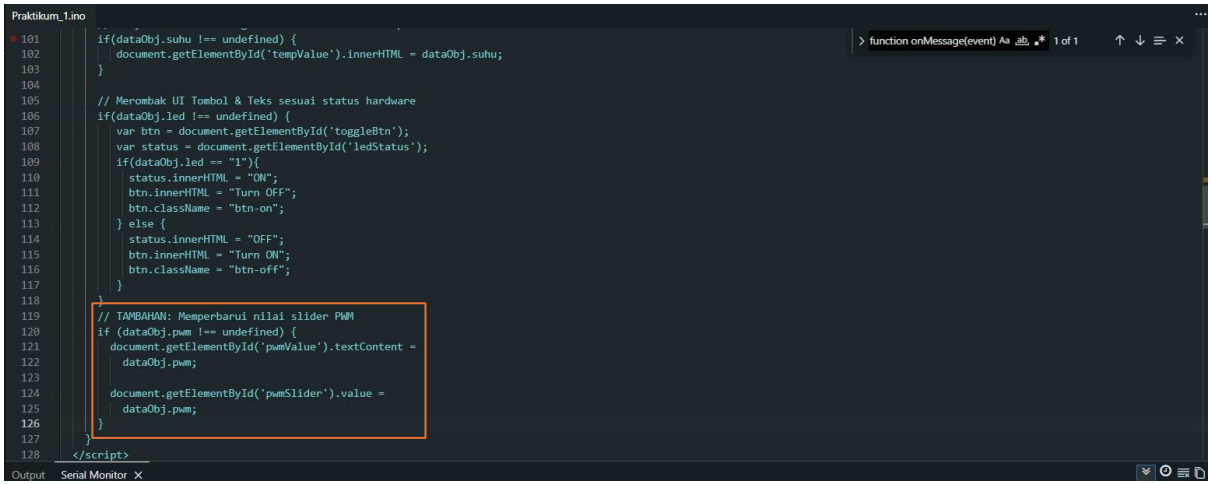
    if (*endPtr == '\0' &&
        endPtr != message + 4 &&
        value >= 0 && value <= 1023) {

      pwmValue = (int)value;

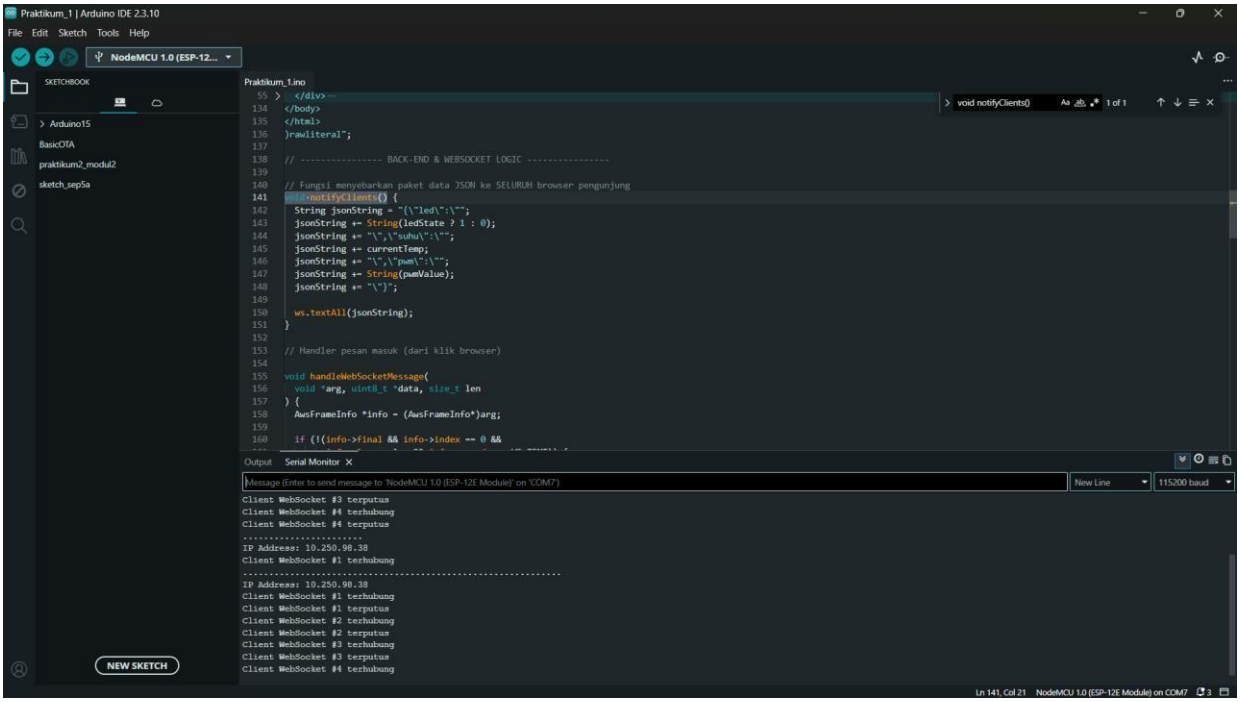
      analogWrite(pwmLedPin, pwmValue);

      notifyClients();
    }
  }
}
```

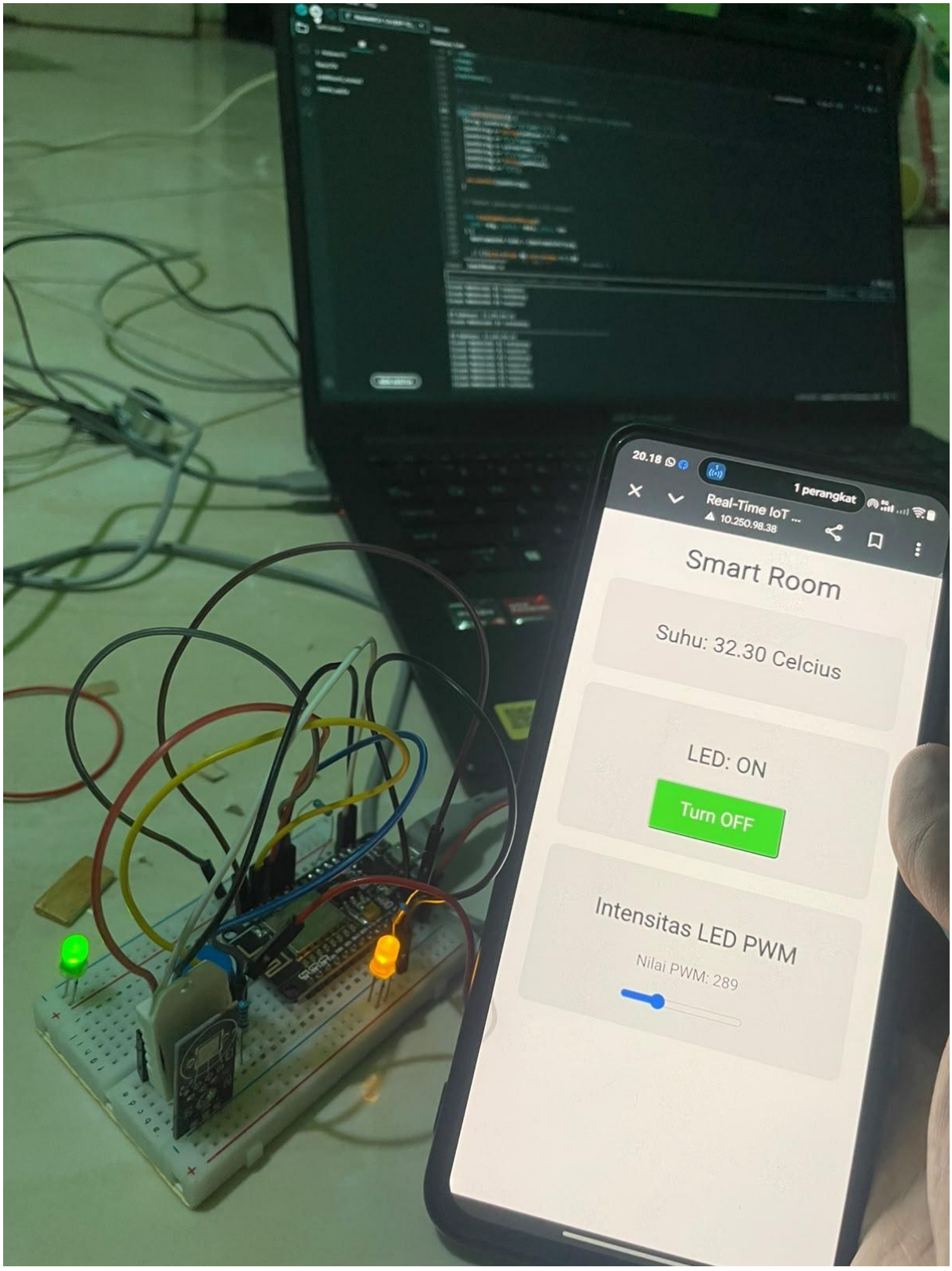
7. Tambahkan pembaruan slider dari WebSocket



Output Ketika di running di serial

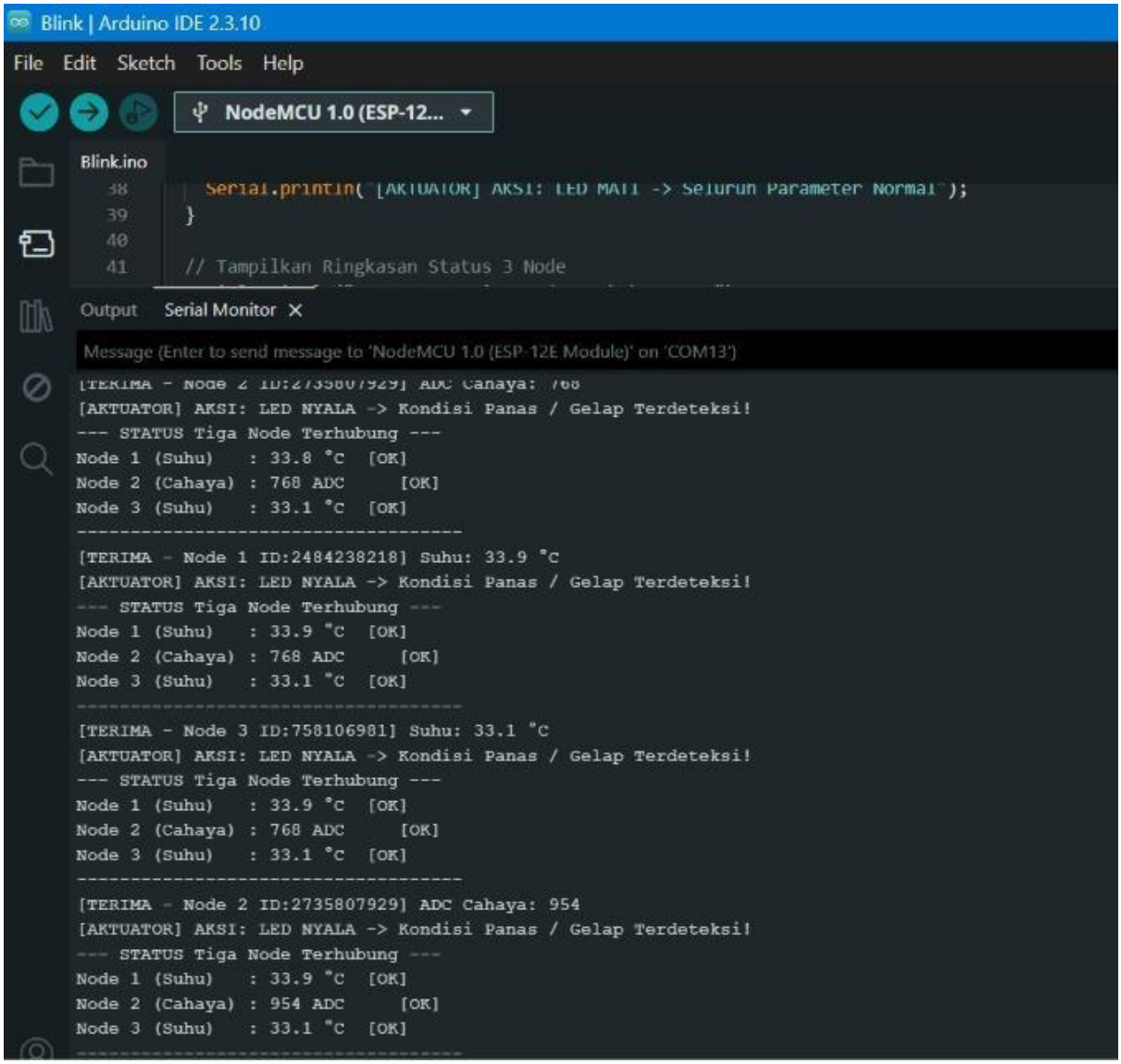


Bukti telah berhasil :

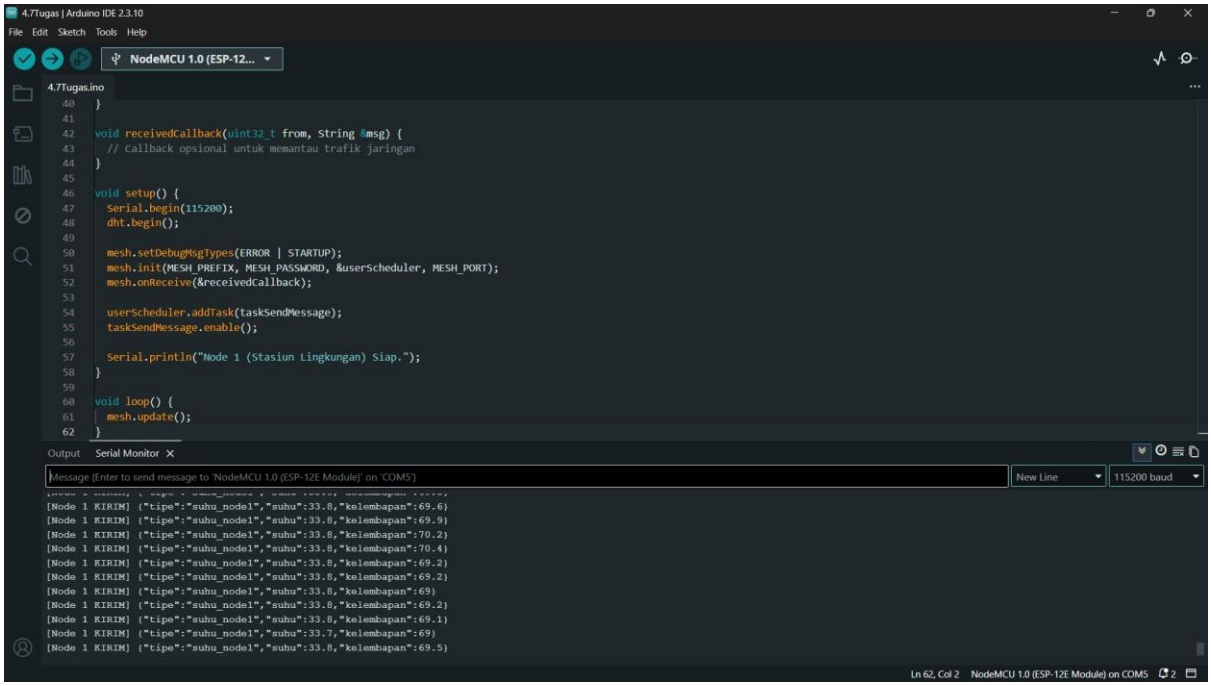


Tugas Modul 5

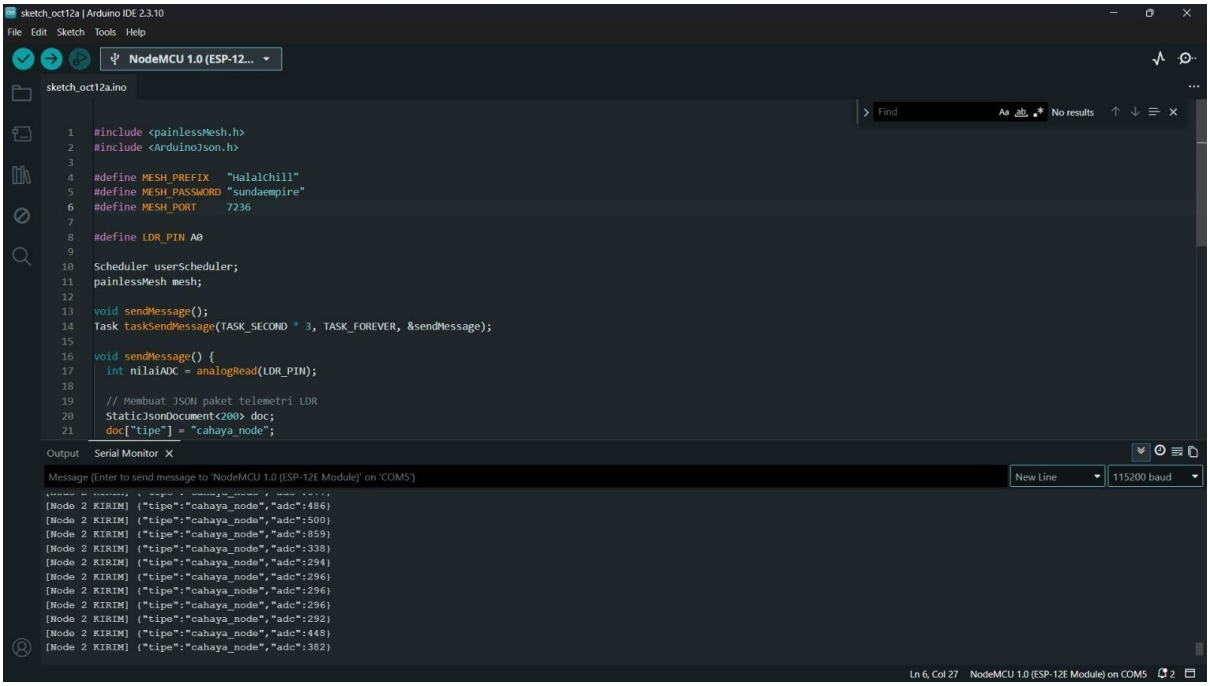
Aktuator



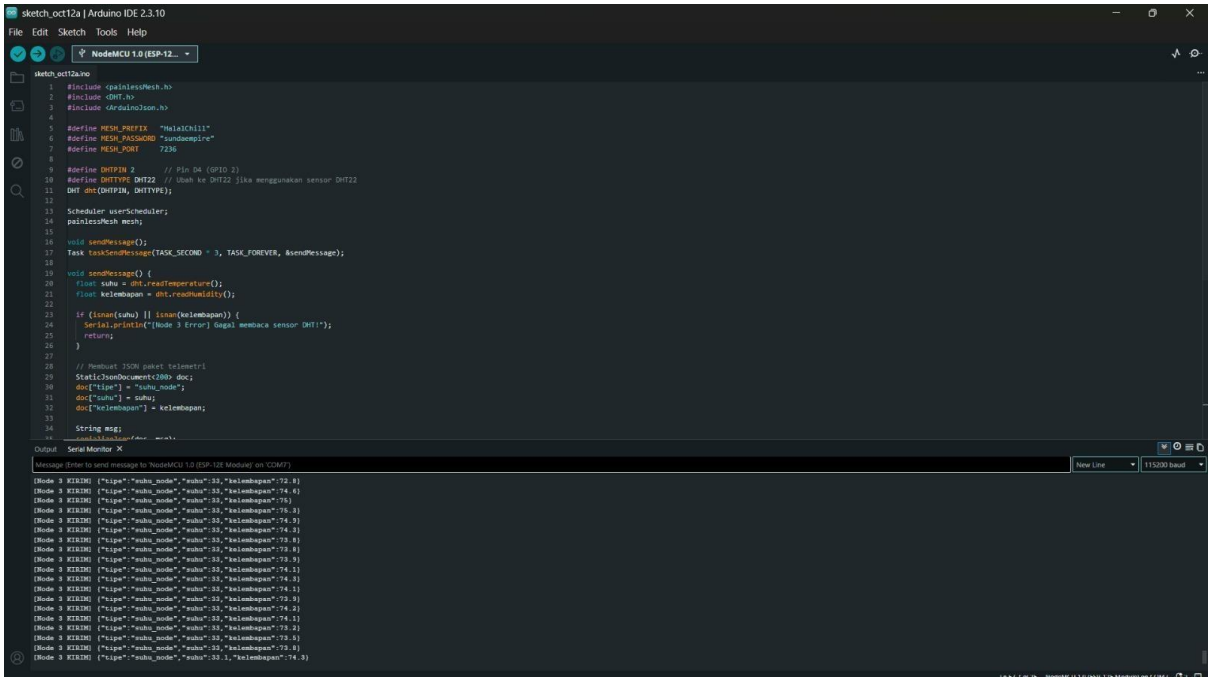
Node 1


4.7Tugas | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
4.7Tugas.ino
}
void receivedCallback(uint32_t from, String &msg) {
// Callback opsional untuk memantau trafik jaringan
}
void setup() {
Serial.begin(115200);
dht.begin();
mesh.setDebugMsgTypes(ERROR | STARTUP);
mesh.init(MESH_PREFIX, MESH_PASSWORD, &userScheduler, MESH_PORT);
mesh.onReceive(&receivedCallback);
userScheduler.addTask(taskSendMessage);
taskSendMessage.enable();
Serial.println("Node 1 (Stasiun Lingkungan) Siap.");
}
void loop() {
mesh.update();
}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.6}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.9}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":70.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":70.4}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.2}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.1}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.7,"kelembapan":69}
[Node 1 KIRIM] {"tipe":"suhu_node1","suhu":33.8,"kelembapan":69.5}
Ln 62, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM5

Node 2


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <painlessMesh.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "Halalchill"
#define MESH_PASSWORD "sundaempire"
#define MESH_PORT 7236
#define LDR_PIN A0
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
int nilaiADC = analogRead(LDR_PIN);
// Membuat JSON paket telemetri LDR
StaticJsonDocument<200> doc;
doc["tipe"] = "cahaya_node";
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":486}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":500}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":859}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":338}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":294}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":292}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":448}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":382}
Ln 6, Col 27 NodeMCU 1.0 (ESP-12E Module) on COM5

Node 3


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <painlessMesh.h>
#include <DHT.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "HalaiChili"
#define MESH_PASSWORD "sundaempire"
#define MESH_PORT 7236
#define DHTPIN 2 // Pin D4 (GPIO 2)
#define DHTTYPE DHT22 // Ubah ke DHT22 jika menggunakan sensor DHT22
DHT dht(DHTPIN, DHTTYPE);
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
float suhu = dht.readTemperature();
float kelembapan = dht.readHumidity();
if (isnan(suhu) || isnan(kelembapan)) {
Serial.println("[Node 3 Error] Gagal membaca sensor DHT!");
return;
}
// Membuat JSON paket telemetri
StaticJsonDocument<200> doc;
doc["tipe"] = "suhu_node";
doc["suhu"] = suhu;
doc["kelembapan"] = kelembapan;
String msg;
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')
New Line
115200 baud
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":72.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.6}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":76}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":76.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.2}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.2}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.5}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33.1,"kelembapan":74.3}